Set Theory/Proofs

1. CB 1.2
2. CB 1.3bc
3. Let $A_n = (\frac{1}{2n}, \frac{1}{n})$. Describe the sets

$$A = \bigcap_{n=1}^{\infty} A_n \qquad \text{and} \qquad B = \bigcup_{n=1}^{\infty} A_n.$$

σ-fields/algebras

4. CB 1.11
5. Is the union of two sigma algebras a sigma algebra? Give a proof or counter example.

Probability

6. CB 1.4
7. Let P be a probability function and let A and B be any sets in $\mathcal{B}$. Prove the following:
   (a) $P(B \cap A^c) = P(B) - P(A \cap B)$ (*Hint:* Start with CB 1.2b)
   (b) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (*Hint:* Start with CB 1.2d)
   (c) If $A \subset B$, then $P(A) \leq P(B)$
8. CB 1.13
9. CB 1.7
10. Two bankers each arrive at the station at some random time between 5PM and 6PM (arrival time for each of them is uniformly distributed). If they arrive before 5:50, they stay exactly 10 minutes and then leave. If a banker arrives after 5:50, they will leave at 6 PM. What is the probability that they will be at the station together on a given day?
11. CB 1.18
12. CB 1.22
13. Consider choosing a five-card poker hand from a standard deck of 52 playing cards. (See example 1.2.18 in textbook.) What is the probability of a full house (three cards of the same rank and two cards of another rank)?

Conditional Probability

14. CB 1.34
15. CB 1.35
16. CB 1.36
17. CB 1.37

Problems labeled *CB* can be found in our textbook, *Statistical Inference* by **C**asella & **B**erger.